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$$\begin{aligned} & \int \frac{(2x+1)^2 \sqrt{2x+1}}{\sqrt[3]{(2x+1)^2}} dx \\ &= \int (2x+1)^{2+\frac{1}{2}-\frac{2}{3}} dx \\ &= \int (2x+1)^{\frac{11}{6}} dx \end{aligned}$$

stel $u = 2x + 1$

$$du = 2dx$$

$$\begin{aligned} &= \frac{1}{2} \int u^{\frac{11}{6}} du \\ &= \frac{1}{2} \cdot \frac{u^{\frac{17}{6}}}{\frac{17}{6}} \\ &= \frac{3u^{\frac{17}{6}}}{17} \\ &= \frac{3}{17} (2x+1)^{\frac{17}{6}} + C \end{aligned}$$

References